

Caffeine Improves Simulated 800-m Run Performance without Affecting Severe Exercise-Induced Arterial Hypoxemia

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ABSTRACT

DOBASHI, K., Y. FUKAZAWA, A. KATAGIRI, Y. ENOMOTO, T. NISHIYASU, and N. FUJII. Caffeine Improves Simulated 800-m Run Performance without Affecting Severe Exercise-Induced Arterial Hypoxemia. *Med. Sci. Sports Exerc.*, Vol. 56, No. 2, pp. 350–361, 2024. **Purpose:** Although caffeine is known to possess ergogenic effects, previous studies demonstrated no effect of caffeine on 800-m run performance outdoors, which might be due to several uncontrolled factors including pacing strategies. We hypothesized that caffeine ingestion improves a pace-controlled simulated 800-m run performance. We also hypothesized that exercise-induced arterial hypoxemia occurs during the simulated 800-m run, and this response is mitigated by caffeine-induced increases in exercise ventilation. **Methods:** In a randomized, double-blind, placebo-controlled and crossover design, 16 (3 females) college middle-distance runners who have 800-m seasonal best of 119.97 ± 7.64 s ingested either 1) placebo (6 mg of glucose per kilogram of body weight) or caffeine (6 mg of caffeine per kilogram of body weight). Then they performed an 800-m run consisting of 30-s running at 103% of their 800-m seasonal best, followed by running at 98% of seasonal best until exhaustion, which mimics actual 800-m run pacing pattern. **Results:** Running time to exhaustion was extended by $7.3\% \pm 6.2\%$ in the caffeine-ingested relative to placebo trial (123 ± 12 vs 114 ± 9 s, $P = 0.04$). Arterial oxygen saturation markedly decreased during the simulating running, but this response was similar ($76.6\% \pm 5.7\%$ vs $81.1\% \pm 5.2\%$, at 113 s of the simulating running) between the caffeine and placebo trials ($P \geq 0.23$ for time-supplement interaction and main effect of supplement). Minute ventilation, oxygen uptake (all $P \geq 0.36$ for time-supplement interaction and main effect of supplement), and rate of perceived exertion (all $P \geq 0.11$) did not differ between the trials throughout the simulating running. Heart rate was higher in the caffeine-ingested trial throughout the simulated running ($P < 0.01$ for main effect of supplement). Postexercise blood lactate concentration was higher in the caffeine trial ($P = 0.02$). **Conclusions:** Caffeine ingestion improves simulated 800-m run performance without affecting exercise ventilation and severe exercise-induced arterial hypoxemia. **Key Words:** ERGOGENIC AID, ANAEROBIC METABOLISM, AEROBIC METABOLISM, EXERCISE-INDUCED ARTERIAL HYPOXEMIA

Caffeine (1,3,7-trimethylxanthine) is widely used as an ergogenic aid to improve several aspects of exercise performance, such as prolonged (e.g., 30-min) cycling (1), muscle strength (2), and short-term high-intensity cycling (e.g., 30-s Wingate test) (2). Caffeine or coffee containing caffeine also improves running performance including 100 m (3), 1000 m (4), 1500 m (5), 1 mile (6), and 5 km (7). However, Marques et al. (8) reported that ingestion of coffee containing

5.5 mg·kg⁻¹ of body weight of caffeine did not improve 800-m run performance relative to decaffeinated coffee ingestion in amateur endurance runners. Similarly, a subsequent study by Ramos-Campo et al. (9) reported that 6 mg·kg⁻¹ of body weight of caffeine ingestion did not improve 800-m run performance relative to placebo in trained runners. Therefore, current evidence does not support ergogenic effect of caffeine on 800-m run performance.

However, it may be premature to conclude that caffeine does not improve 800-m run performance based on the aforementioned previous studies (8,9). For example, the two studies used 800-m time trial at track outdoors. Hence, their results may be confounded by different ambient conditions and/or pacing patterns between trials. As for the latter, better 800-m run performance seems to be achieved by faster time in initial 200 m (10,11), highlighting the importance of pacing strategy. Controlling environmental conditions and pacing in the laboratory setting is required to judge if caffeine intake has a potential to improve 800-m run performance. It is also important to highlight that the aforementioned study by Marques et al. (8) assessed the effect of caffeine by comparing caffeinated versus

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TABLE 1. Characteristics of the participants.

	Age, yr	Height, m	Body Mass, kg	800-m Seasonal Best, s	800-m Personal Best, s	IAAF Score of 800-m Seasonal Best	$\dot{V}O_{2\max}$, mL·kg ⁻¹ ·min ⁻¹
Male (<i>n</i> = 13)	21 ± 2	1.75 ± 0.05	61.9 ± 4.7	117.10 ± 4.35	116.50 ± 3.79	837 ± 112	65.4 ± 6.8
Female (<i>n</i> = 3)	20 ± 1	1.57 ± 0.03	45.3 ± 5.1	132.41 ± 6.24	132.41 ± 6.24	952 ± 100	56.7 ± 4.9
Combined (<i>n</i> = 16)	20 ± 2	1.72 ± 0.09	58.8 ± 8.1	119.97 ± 7.65	119.48 ± 7.60	859 ± 117	63.8 ± 7.3

Values expressed in mean ± SD.

noncaffeinated coffee. Coffee contains chlorogenic acid that may blunt the caffeine's ergogenic aid (12). Consequently, the effect of caffeine assessed in the studies by Marques et al. (8) and Ramos-Campo et al. (9) might be underestimated.

Caffeine's ergogenic effects can be explained by increasing central nervous activity (13) via its inhibitory action on adenosine receptors in the brain (14), which can induce greater recruitment of motor units (15,16) and reductions in rating of perceived exertion (RPE) (17,18). Caffeine may also increase the rate of calcium release from the sarcoplasmic reticulum (19), augmenting muscle contraction. In addition, caffeine possesses the capability to enhance exercise ventilation, potentially through adenosine-dependent mechanisms (20), at given workloads or running speeds, spanning from submaximal (21) to maximal (22,23) exercise. This caffeine-induced increase in ventilation could improve the performance of the 800-m run; in this context, sustained high-intensity exercise in a normoxic environment can lead to 3% to 4% reductions in arterial O₂ saturation (SpO₂) from rest (so-called exercise-induced arterial hypoxemia (EIAH)) (24,25). EIAH is thought to impair exercise performance (24), possibly by diminishing oxygen supply to cerebral tissues and active muscles, thus leading to central (26) and peripheral (27) fatigue, respectively. Chapman and Stager (23) demonstrated that caffeine ingestion increased exercise ventilation during the same maximal intensity of treadmill running, partially mitigating EIAH; however, it remains unknown if EIAH occurs in 800-m running in trained runners and if caffeine increases exercise ventilation, minimizing EIAH, thereby improving 800-m run performance.

In the present study, we hypothesized that caffeine ingestion improves pace-controlled simulated 800-m run performance by increasing ventilation, thereby reducing EIAH.

METHODS

Ethical approval. This study was approved by the Human Subjects Committee of the University of Tsukuba (No. 020-182), in accordance with the Declaration of Helsinki, except for registration in a database. All participants provided written informed consent before participating in this study.

Participants. Healthy 13 male and 3 female college middle-distance runners participated in the present study. Information pertaining to the physical characteristics, 800-m seasonal best time (male: 108.33–125.27 s; female: 127.83–139.52 s), 800-m personal best time (male: 108.33–120.80 s; female: 127.83–139.52 s), International Association of Athletics Federations (IAAF) score (28) calculated by seasonal best, and maximal oxygen uptake ($\dot{V}O_{2\max}$) are presented in Table 1. One male and two female participants had taken part in interuniversity

track and field championships. Eight participants (1 female) consume caffeine everyday (15–150 mg·d⁻¹), whereas the others do not consume caffeine daily. We estimated that a minimum sample size of 12 needed to detect a significant difference in exercise performance evaluated by time to exhaustion during high-intensity exercise, using the calculated effect size of *d* = 0.92 (29), with 80% power and a significance level of 0.05.

None of the participants were smokers or taking prescription medications. All participants belonged to the same university middle distance team and regularly trained (120–240 min of structured training per day, 6 days·wk⁻¹). All refrained from caffeine and alcohol for ≥24 h and foods for ≥2 h before each experimental trial. They were also instructed to avoid intense exercise the night before each trial. No female participants were on contraceptives. We did not control menstrual cycle for the female participants because a previous work reported that the menstrual cycle does not affect ventilation, metabolic rate, perceived exertion, and exercise performance under temperate conditions (30).

Preliminary session. Before the experimental session, all participants completed an incremental running test on a treadmill (ORK-7000; OHTAKE-Root Kogyo, Iwate, Japan) at a 1% grade (31) to determine their $\dot{V}O_{2\max}$. The initial velocity was set to 190, 210, or 230 m·min⁻¹, depending on the participant's running ability, and was increased by 20 m·min⁻¹ each stage up to a total of five or six stages. Each stage consisted of 3-min running interspaced by 2-min rest. After completion of five or six stages, participants rested for 3 min, then they started an incremental running test with a running speed set to that used for the initial stage noted above. The treadmill velocity was increased by 10 m·min⁻¹ every minute until exhaustion. The termination of exercise was determined when two of the following criteria were met: 1) age-predicted maximal heart rate (HR) was achieved (i.e., 220 – age), 2) respiratory exchange ratio (RER) exceeded 1.15, and/or 3) blood lactate concentration exceeded 8.00 mmol·L⁻¹ (32). Expired volume and gases were analyzed continuously using a metabolic cart (AE310-s; Minato Medical Science, Osaka, Japan). $\dot{V}O_{2\max}$ was determined as the highest $\dot{V}O_2$ recorded over a 30-s period. One to 2 months after the test, the participants joined an experimental session described hereinafter.

Experimental session. An overview of the experimental session is depicted in Figure 1. The experimental session was performed in an environmental chamber (Shimazu, Kyoto, Japan) regulated to ~25°C. Upon arrival at the laboratory, body mass was measured by a weight scale platform (model ICS429; Mettler Toledo, Greifensee, Switzerland). After instrumentation, the participants rested in a sitting position for 15 min and then ingested placebo (6 mg of glucose per

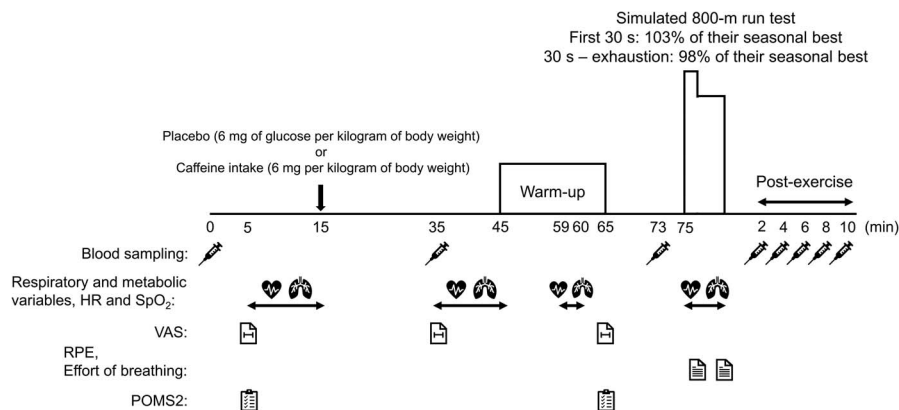


FIGURE 1—Experimental protocol. VAS, visual analog scale. Warm-up was consisted of 5-min stretching, 10-min submaximal run ($199 \pm 1 \text{ m} \cdot \text{min}^{-1}$), 90-s recovery, and $2 \times 15\text{-s}$ sprinting ($386 \pm 14 \text{ m} \cdot \text{min}^{-1}$) interspaced with 3-min recovery. The velocities during submaximal and high-intensity runs were chosen by each participant and were identical between the placebo and caffeine trials within a participant.

kilogram of body weight in capsules) or caffeine (6 mg of caffeine per kilogram of body weight in capsules) in a randomized, double-blind, and crossover procedure. The dose of caffeine used has been shown to increase exercise minute ventilation ($\dot{V}_E$) (21,22) and exercise performance (33). The two trials were separated by at least 6 days. Nine participants started the experiment in the morning (8:30–9:30 AM), whereas the others started the experiment in the afternoon (1:30–5:30 PM), depending on their schedules. The time of day for the experiment was matched between the two trials within a participant. The preliminary and main experiments were conducted between August and November in 2021. Thirty minutes after ingesting placebo or caffeine, the participants performed a warm-up consisted of stretching (5-min), 10-min run at submaximal speed ($199 \pm 1 \text{ m} \cdot \text{min}^{-1}$), 90-s resting, and two bouts of 15-s high-intensity runs ($386 \pm 14 \text{ m} \cdot \text{min}^{-1}$) interspaced with a 3-min recovery. The velocities during the submaximal and high-intensity runs were determined based on each runner's preference. All activities during the warm-up were matched between the two trials within a participant.

After completion of the warm-up, the participants rested on a treadmill for 10 min. Then they performed a simulated 800-m run test consisting of 30-s running at speed of 103% of their 800-m seasonal best ($413 \pm 12 \text{ m} \cdot \text{min}^{-1}$), followed by running at speed of 98% of their seasonal best ($373 \pm 11 \text{ m} \cdot \text{min}^{-1}$) until exhaustion. The initial 30-s running speed was set above averaged 800-m seasonal best speed, as the initial 200-m of 800-m race is typically faster than the rest of pace (10,11). Based on our preliminary experiment, we identified that the pacing pattern used in the main experiment enables middle-distance runners to run for ~ 2 min, which closely approximates the duration of 800-m race. We elected to use a biphasic constant velocity running protocol, owing to the limitation of our treadmill, which does not allow instantaneous and precise modulation of running velocity throughout the 800-m run simulation. After the simulated 800-m run test, participants sat in a chair next to the treadmill for 10 min.

Measurements. Time to exhaustion during the simulated 800-m run test was measured as exercise performance. Expired volume and O_2 and CO_2 concentrations were analyzed

using the metabolic cart (AE310-s) similar to the preliminary session. The flow sensor was calibrated using a syringe that blew a fixed volume (2 L) of air. The O_2 and CO_2 sensors were calibrated with reference gases of known concentration (O_2 : 14.90%, CO_2 : 5.05%, N_2 balanced). $\dot{V}\text{O}_2$, carbon dioxide elimination ($\dot{V}\text{CO}_2$), $\dot{V}_E$, tidal volume (V_T), respiratory frequency (f_R), RER, and ventilatory equivalent for $\dot{V}\text{O}_2$ ($\dot{V}_E/\dot{V}\text{O}_2$), and ventilatory equivalent for $\dot{V}\text{CO}_2$ ($\dot{V}_E/\dot{V}\text{CO}_2$) were recorded on a breath-by-breath basis. HR was recorded using an HR monitor (V-800; Polar, Kempele, Finland). SpO_2 was measured from the participant's forehead using a pulse oximeter (PM10N; Nellcor, Brampton, Ontario, Canada). Fingertip blood samples were collected from warmed finger based on which blood lactate concentrations were assessed in duplicate using a lactate analyzer (Lactate Pro 2; Arkay, Kyoto, Japan). RPE for the whole body (6–20 scale) or leg (0–10 scale), and effort of breathing (0–10 scales) (34) were assessed at the initial 30 s and end of simulated 800-m run test. Participants memorized RPE and effort of breathing during running and reported them immediately after the simulated 800-m running test. We did not document RPE and effort of breathing during running to mitigate disturbances that might divert the participants' attention from the high-intensity running task. Moreover, soliciting participants to report their RPE and effort of breathing could pose a potential hazard, as evidenced in our preliminary experiment where participants experienced instability and near falls on the treadmill due to the high-speed running. Readiness to perform the simulated 800-m running test was assessed at 10 min before ingestion of placebo or caffeine, 10 min before the warm-up, and 10 min before the simulated 800-m running test using a visual analog scale based on previous work (35). To assess if caffeine intake negatively influences sleep quality as reported in a previous study (9), next morning the participants completed a questionnaire consisting of five questions (36) (0–5 scales for each): 1) How have you slept? 2) Did you have difficulty falling asleep? 3) Was your sleep restless? 4) Did you wake up prematurely and were not able to get back to sleep? and 5) How many times did you wake up?). Sleep score was calculated as follows.

Sleep score = 1) + 2) + 3) + 4) – 5)

Total score ranges from –5 to 20 scales, where –5 is defined as low sleep quality and 20 is defined as high sleep quality.

Profile of Mood States Second Edition (POMS2) (37) was assessed 10 min before capsule ingestion and 10 min before the simulated 800-m run test.

Data analyses. The highest blood lactate concentration obtained during the postexercise recovery periods after the simulated 800-m run was used as peak blood lactate concentration. Resting respiratory and metabolic variables, HR, and SpO₂ at preingestion or postingestion were assessed by averaging data over the 10 min before ingestion of placebo or caffeine (5–15 min in Fig. 1) or the last 10 min of pre-warm-up resting period (35–45 min in Fig. 1). The values during warm-up were assessed using average data over a period extending from the 9th to the 10th of the 10 min of submaximal running (59–60 min in Fig. 1). The values during the simulated 800-m run were assessed by averaging data every 10 s and the last 10 s of the simulated 800-m run. Furthermore, the value during the final 10 s of the simulated 800-m run in one trial, in which participants ran for a shorter duration, was assessed and compared with the value at the corresponding time point in the other trial (i.e., 113 ± 9 s). Because of technical difficulties, HR for one female participant and SpO₂ for one male participant during preingestion, postingestion, and submaximal warm-up were not acquired. In addition, HR for two male and one female participants and SpO₂ for one male and one female participants during high-intensity 800-m run, and POMS2 for one female participant were not acquired.

Statistical analyses. Normality of variables was verified by the Shapiro–Wilk test and Q–Q plot; the latter was checked by two investigators. Student’s paired *t*-test was used for between-trial differences in time to exhaustion; peak blood lactate concentration; RPE measured at the initial 30 s and end of 800-m simulated run; readiness assessed at preingestion, postingestion, and pre-800-m run; sleep score; and POMS2 variables at preingestion and pre-800-m run. Student’s paired *t*-test was also used for between-trial differences for respiratory and metabolic variables, HR, and SpO₂ at preingestion and postingestion under resting states and during submaximal running warm-up and end of 800-m run. On the other hand, the Shapiro–Wilk test and Q–Q plot revealed that individual sleep scores (1–5) and some variables for POMS were not normally distributed. Therefore, the Wilcoxon test was used for between-trial differences.

Respiratory and metabolic variables, HR, and SpO₂ during the simulated 800-m run test were analyzed by two-way repeated-measures ANOVA with two factors of the supplement (levels: placebo, caffeine) and time (levels: 10, 20, 30, 40, 50, 60, 70, 80, and 113 s). For time-dependent changes in blood lactate concentration, we also used two-way ANOVA with factors of supplement (levels: placebo, caffeine) and time (preingestion, postingestion, pre-800-m run, every 2 min after the simulated 800-m run test). When the interaction was detected for the two-way ANOVA, *post hoc* multiple comparisons were performed with the Bonferroni correction. All of

the data used for ANOVA and *t*-test were normally distributed as verified by the Shapiro–Wilk test, with very few exceptions (e.g., SpO₂ in the placebo trial during the simulated 800-m run test). For these variables, we confirmed a minor deviation from normality through visual inspection with Q–Q plots. Given that ANOVA is a robust test against the normality assumption, thus including just a very few nonnormally distributed data in the ANOVA is acceptable.

The Pearson’s correlation analysis was performed on the relationship between change (Δ) in running time from caffeine to placebo trials and IAAF score of their 800-m seasonal best, and the relationship between the change in running time and change in SpO₂, $\dot{V}O_2$, $\dot{V}_E$, or peak blood lactate concentration from caffeine to placebo trials during the 800-m run. Effect size for pairwise comparisons was calculated using Cohen’s *d* (0.01 = very small, 0.2 = small, 0.5 = medium, 0.8 = large, 1.2 = very large, 2.0 = huge) (38). Statistical significance was set at $P < 0.05$. All statistical analyses were performed using GraphPad Prism, v.8.4.3 (GraphPad Software Inc., San Diego, CA).

Data are presented as the mean ± 95% confidence interval when normally distributed, whereas median and range are used when not normally distributed. Participants’ characteristics are presented in mean ± SD.

RESULTS

Before, during, and after simulated 800-m run test.

Time to exhaustion during the simulated 800-m run was extended by 7.3% ± 6.2% in the caffeine relative to the placebo trial (123 ± 12 vs 114 ± 9 s, $d = 0.38$; Fig. 2A). Peak blood lactate concentration was higher in the caffeine than in the placebo trial (19.1 ± 1.4 vs 17.6 ± 1.0 mmol·L^{–1}, $d = 0.65$; Fig. 2B). Although no time–supplement interaction was detected, a main effect of supplement was observed for HR such that HR was higher in the caffeine-ingested trial relative to the placebo trial throughout the simulated 800-m run (Fig. 3A). SpO₂ decreased with exercise time in both trials ($P < 0.01$ for a main effect of time), reaching 76.6% ± 5.7% in the caffeine treated trial and 81.1% ± 5.2% in the placebo trial at 113 s of simulating running; however, neither the interaction between time and supplement nor a main effect of supplement was observed (Fig. 3B). Along this line, neither time–supplement interaction nor a main effect of supplement was detected on $\dot{V}_E$ and the other respiratory and metabolic variables (Figs. 4A–F) except $\dot{V}_E/\dot{V}O_2$ and $\dot{V}_E/\dot{V}CO_2$ wherein the interaction was significant (Figs. 4G, H). At 10 s into the exercise, $\dot{V}_E/\dot{V}CO_2$ was higher in the caffeine than the placebo trial (Fig. 4H, $P = 0.02$). Data showing between-supplement comparisons at end of 800-m run are provided in Table 2 wherein RER and HR were higher in the caffeine-ingestion conditions.

RPE for whole body at the initial 30 s (14.6 ± 0.6 vs 14.3 ± 0.7, a.u., $d = 0.28$) and end (19.4 ± 0.4 vs 19.1 ± 0.4, a.u., $d = 0.37$) of simulated 800-m run test was similar between the placebo and caffeine trials (Fig. 5A). RPE for leg at the initial 30 s (5.6 ± 0.8 vs 5.0 ± 0.5, a.u., $d = 0.42$) and end (9.4 ± 0.4 vs 9.4 ± 0.4, a.u., $d < 0.01$) of simulated 800-m run test was also comparable

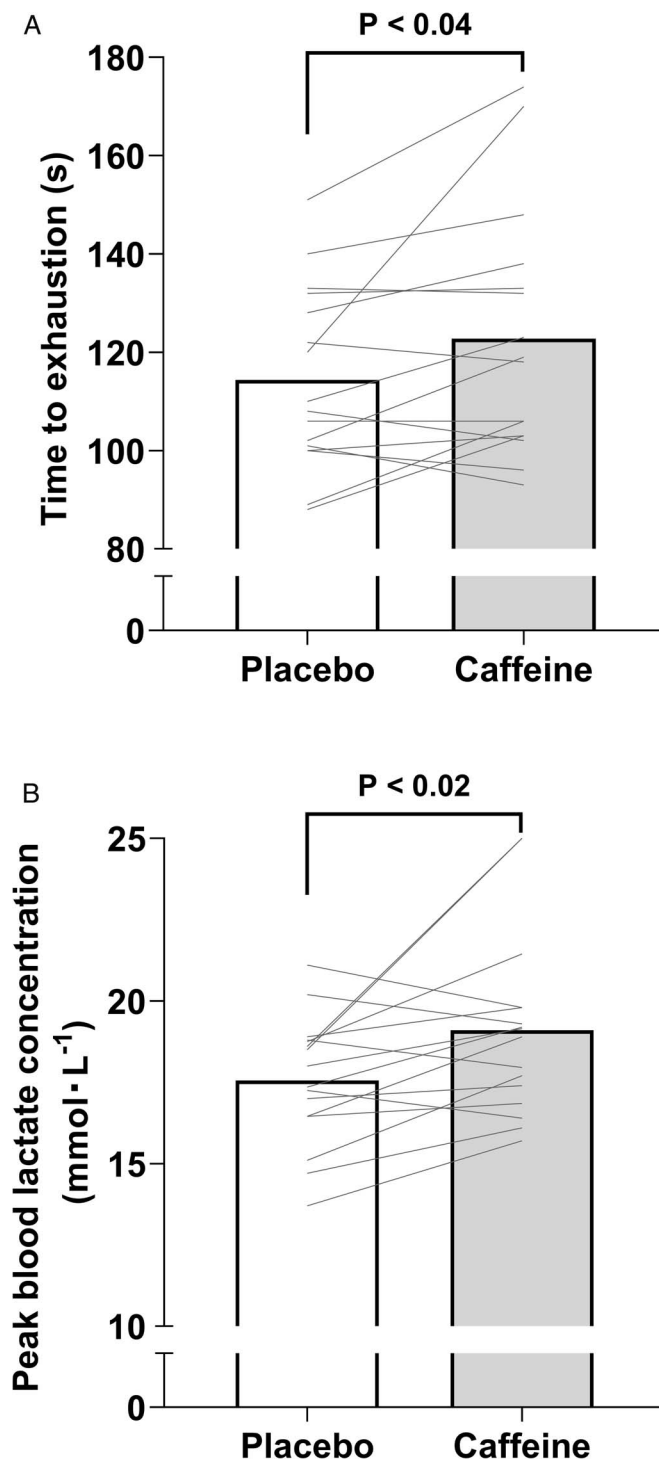


FIGURE 2—Time to exhaustion during a simulated 800-m run test (A) and peak blood lactate concentration observed after a simulated 800-m run (B) in both the placebo and caffeine trials. Both individual data and means are presented ($n = 16$). Data were analyzed using a paired t -test.

between the trials (Fig. 5B). In addition, effort of breathing at the initial 30 s (4.9 ± 0.9 vs 4.3 ± 0.6 a.u., $d = 0.39$) and end (8.6 ± 0.8 vs 8.4 ± 0.5 a.u., $d = 0.15$) of simulated 800-m run test did not differ between the trials (Fig. 5C). Readiness to perform the simulated 800-m running test was not different between the placebo and caffeine ingestion trials at preingestion (49 ± 12 vs 44 ± 10 a.u., $d = 0.21$), postingestion (56 ± 9 vs 53 ± 11 a.u., $d = 0.16$), and pre-800-m run (79 ± 7 vs 79 ± 10 a.u., $d = 0.01$) (Fig. 5D).

Statistical outcomes for main variables remained similar even when excluding female data (e.g., time to exhaustion was longer in the caffeine vs placebo trials, $d = 0.45$, $P = 0.03$, Supplemental Fig. 1A, Supplemental Digital Content, <http://links.lww.com/MSS/C910>; peak blood lactate concentration was greater in the caffeine vs placebo trials, $d = 0.75$, $P < 0.02$, Supplemental Fig. 1B, Supplemental Digital Content, <http://links.lww.com/MSS/C910>; no between-trial differences in $\dot{V}_E$, $\dot{V}O_2$, and SpO_2

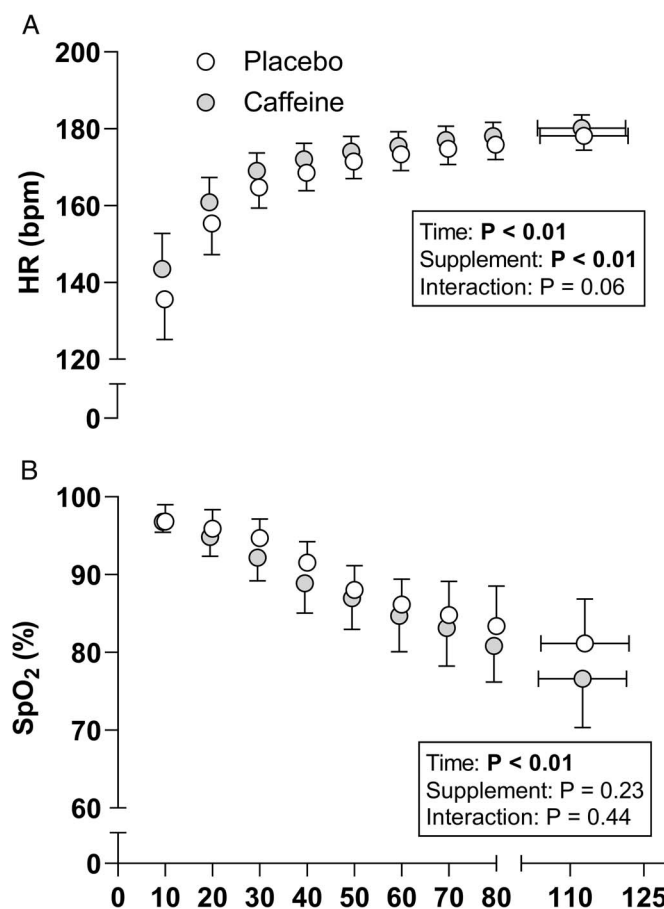


FIGURE 3—Time-dependent changes in HR (A; $n = 13$), and SpO₂ (B; $n = 14$) during a simulated 800-m run test in both the placebo and caffeine trials. The value during the final 10 s of the simulated 800-m run in one trial, in which participants ran for a shorter duration, was assessed and compared with the value at the corresponding time point in the other trial (i.e., 113 $\pm$ 9 s). Data were analyzed using two-way repeated-measures ANOVA, followed by Bonferroni correction. Data are presented as mean $\pm$ 95% confidence interval.

presented in Supplemental Fig. 2, Supplemental Digital Content, <http://links.lww.com/MSS/C910>). Change in time to exhaustion from caffeine to placebo trials during the simulated 800-m run did not correlate with any variables assessed (Supplemental Fig. 3, Supplemental Digital Content, <http://links.lww.com/MSS/C910>). Time-course changes in blood lactate concentrations are reported in Supplemental Fig. 4, Supplemental Digital Content, <http://links.lww.com/MSS/C910>. To investigate an order effect, we also analyzed the values between the first and second trials independently of supplement. Neither main effect of supplement nor interaction was detected, except for the readiness measured before and after ingestion (Supplemental Figs. 5–9, Supplemental Digital Content, <http://links.lww.com/MSS/C910>).

During rest and submaximal running warm-up.

Resting $\dot{V}_E$ at postingestion tended to be higher in the caffeine than the placebo trial ($P = 0.06$, $d = 0.28$; Table 2). Along this line, RER and $\dot{V}CO_2$ at postingestion were higher in the caffeine than the placebo trial ($P < 0.01$, $P = 0.03$, $d = 1.02$, and $d = 0.37$, respectively; Table 2). However, between-supplement differences were not observed during submaximal running warm-up (Table 2). Before the simulated 800-m run test, blood lactate concentration was higher in the caffeine (2.6 ± 0.5 mmol·L⁻¹)

than the placebo (1.9 ± 0.5 mmol·L⁻¹) trial ($P < 0.01$, $d = 0.68$; Supplemental Fig. 4, Supplemental Digital Content, <http://links.lww.com/MSS/C910>). Neither POMS scores differed between the trials at preingestion and postingestion (Supplemental Table 2, Supplemental Digital Content, <http://links.lww.com/MSS/C910>).

Subjective sleep quality. Although each sleep score (1–5) was not different between the trials (Supplemental Table 1, Supplemental Digital Content, <http://links.lww.com/MSS/C910>), the calculated score of sleep quality was lower in the caffeine relative to placebo trial (15.9 ± 2.3 vs 16.9 ± 2.0 , a.u., $P < 0.01$, $d = 0.46$; Fig. 6).

DISCUSSION

We investigated the effect of caffeine (6 mg of caffeine per kg of body weight) on exercise performance, SpO₂, and respiratory, metabolic, and cardiac responses during simulated 800-m run in a randomized, double-blind, placebo-controlled, crossover procedure. We found that caffeine ingestion improved the simulated 800-m run performance by $7.3\% \pm 6.2\%$ as per our original hypothesis. Severe EIAH was observed during the simulated 800-m run (SpO₂ of $\sim 80\%$), but this response and exercise $\dot{V}_E$

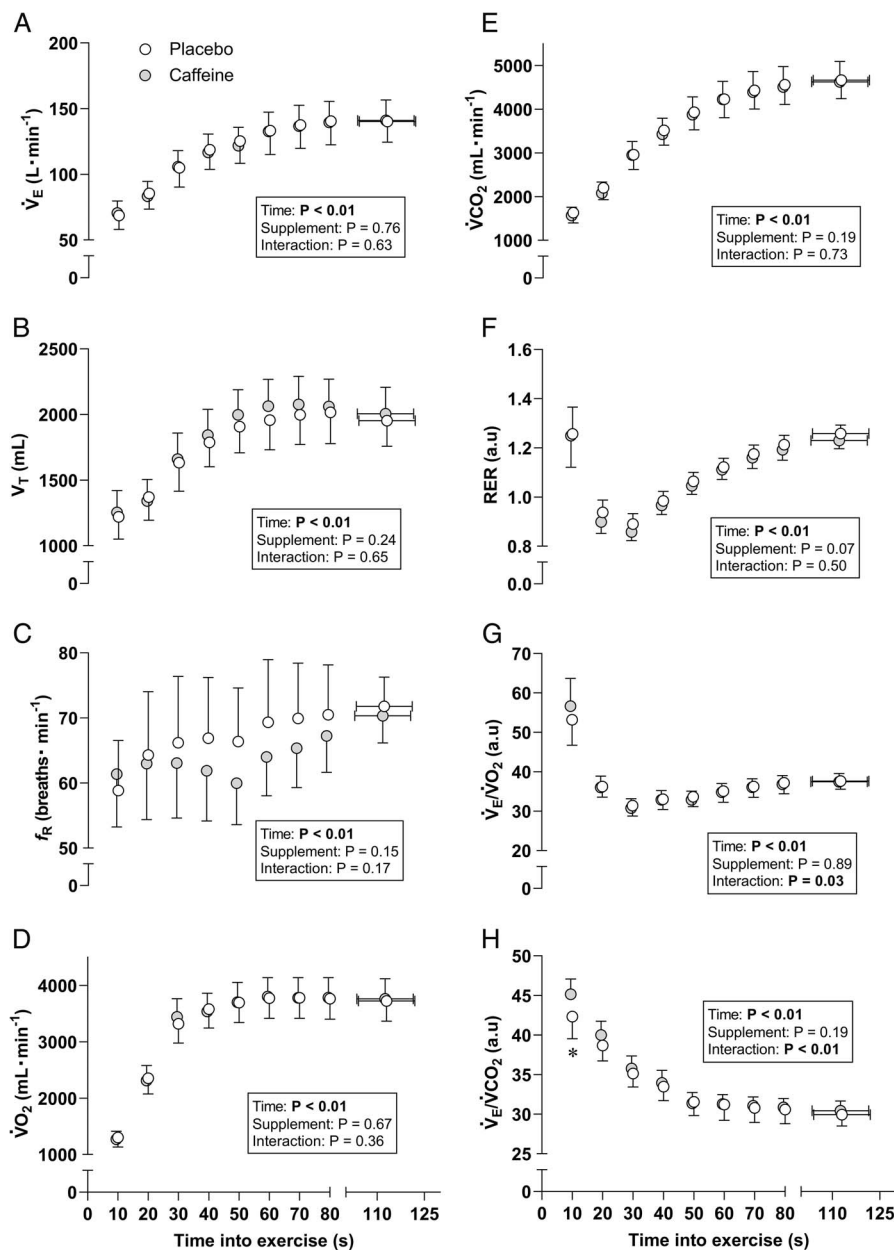


FIGURE 4—Time-dependent changes in $\dot{V}_E$ (A), V_T (B), f_R (C), $\dot{V}O_2$ (D), $\dot{V}CO_2$ (E), RER (F), $\dot{V}_E/\dot{V}O_2$ (G), and $\dot{V}_E/\dot{V}CO_2$ (H) during a simulated 800-m run test in both the placebo and caffeine trials ($n = 16$). The value during the final 10 s of the simulated 800-m run in one trial, in which participants ran for a shorter duration, was assessed and compared with the value at the corresponding time point in the other trial (i.e., 113 ± 9 s). Data were analyzed using a two-way repeated-measures ANOVA, followed by Bonferroni correction. Data are presented as mean $\pm$ 95% confidence interval. * indicates between-trial difference ($P = 0.04$).

were not modulated by caffeine ingestion. $\dot{V}O_2$ and perceptions including RPE were unaffected by caffeine ingestion, but peak blood lactate concentration and exercise HR were increased by caffeine ingestion. These results suggest that caffeine ingestion improves 800-m run performance without influencing severe EIAH, aerobic metabolisms, and perceptions.

We demonstrated that simulated 800-m run performance was improved by caffeine ingestion relative to the placebo trial. Given that the average improvement was not small in magnitude ($7.3\% \pm 6.3\%$), with the observed $d = 0.41$ being close to $d = 0.50$ categorized as “medium,” the effect of caf-

feine on 800-m run performance seems to be meaningful in well-trained middle-distance runners. Contrary to our findings, previous work failed to detect the effect of caffeine ingestion on 800-m time trial at track outdoors (8,9). This discrepancy could be attributed to different experimental protocols used. We used fixed running speed to minimize the influence of pacing strategy on 800-m run performance, whereas previous studies did not fix racing pace during 800-m run (8,9). Unfortunately, the two previous studies did not report details about how 800-m run was conducted (e.g., one runner ran alone or multiple runners ran together), but given that caffeine may alter

TABLE 2. Respiratory, metabolic, and cardiac variables at rest before and after supplementation, during submaximal running warm-up, and at end 800-m run.

	Supplement	Preingestion	P	Postingestion	P	Warm-Up	P	End 800-m Run	P
$\dot{V}_E$, L·min ⁻¹	Placebo	9.4 ± 0.8	0.86	9.9 ± 1.0	0.06	69.3 ± 5.6	0.12	140.3 ± 14.8	0.54
	Caffeine	9.5 ± 1.3		10.6 ± 1.2		70.9 ± 6.0		139.0 ± 14.2	
V_T , mL	Placebo	731 ± 59	0.09	739 ± 83	0.24	1448 ± 138	0.07	1949 ± 185	0.70
	Caffeine	777 ± 54		795 ± 67		1496 ± 134		1963 ± 188	
f_R , breath·min ⁻¹	Placebo	13.3 ± 1.7	<0.05	13.9 ± 1.6	0.55	49.5 ± 5.0	0.34	72.0 ± 4.1	0.14
	Caffeine	12.4 ± 1.9		13.4 ± 1.5		48.5 ± 4.9		70.8 ± 3.5	
$\dot{V}O_2$, mL·min ⁻¹	Placebo	256 ± 21	0.98	273 ± 26	0.52	2541 ± 173	0.73	3717 ± 342	0.92
	Caffeine	256 ± 29		279 ± 31		2551 ± 184		3722 ± 329	
$\dot{V}CO_2$, mL·min ⁻¹	Placebo	220 ± 18	0.44	232 ± 25	0.03	2308 ± 149	0.32	4663 ± 406	0.31
	Caffeine	229 ± 30		254 ± 33		2338 ± 177		4588 ± 433	
RER, a.u.	Placebo	0.86 ± 0.02	0.24	0.85 ± 0.04	<0.01	0.91 ± 0.01	0.38	1.26 ± 0.03	0.03
	Caffeine	0.88 ± 0.03		0.91 ± 0.02		0.92 ± 0.01		1.23 ± 0.03	
$\dot{V}_E/\dot{V}O_2$, a.u.	Placebo	37.1 ± 1.8	0.79	36.6 ± 2.0	0.06	27.3 ± 1.2	0.15	37.8 ± 1.9	0.15
	Caffeine	36.8 ± 1.8		38.0 ± 1.9		27.8 ± 1.3		37.4 ± 2.0	
$\dot{V}_E/\dot{V}CO_2$, a.u.	Placebo	43.0 ± 1.8	0.04	43.1 ± 1.7	0.11	30.0 ± 1.2	0.32	30.6 ± 1.8	0.33
	Caffeine	41.6 ± 1.7		42.0 ± 1.8		30.4 ± 1.4		30.3 ± 1.1	
HR, bpm	Placebo	61 ± 3	0.38	61 ± 3	0.43	146 ± 6	0.39	178 ± 3	0.02
	Caffeine	60 ± 2		59 ± 3		148 ± 6		181 ± 3	
SpO ₂ , %	Placebo	97.7 ± 0.7	0.02	98.0 ± 0.7	0.75	95.2 ± 0.7	0.38	81.1 ± 5.0	0.27
	Caffeine	98.0 ± 0.8		97.8 ± 0.7		94.7 ± 0.8		77.0 ± 6.5	

Values expressed in mean ± 95% confidence interval. The numbers of participants for all variables presented were 16 except HR ($n = 15$) and SpO₂ ($n = 15$). Preingestion and postingestion represent data recorded over the last 10 min of each period. Data during warm-up period were values recorded over a period extending from the 9th to the 10th of the 10 min of warm-up submaximal running (199 ± 1 m·min⁻¹). End 800-m run represents the last 10-s of simulated 800-m run test. Values between the placebo and caffeine were compared by two-tailed *t*-test at each period.

pacing strategy (39), caffeine intake might have altered pacing in the previous studies, negatively affecting 800-m performance, thereby negating any beneficial effect of caffeine. Future study is warranted to delineate whether and how caffeine intake affects pacing strategy during 800-m run.

SpO₂ reached ~80%, the level of which can be categorized as severe EIAH (24) and is comparable to that occurs when running at maximal intensity under moderate hypoxic conditions (e.g., 2500 m above sea level) (40). The SpO₂ of ~80% observed in the present study is much lower than that observed in previous studies wherein 88%–92% of SpO₂ was reported (41–43) in trained individuals running at maximal or near-maximal intensity at sea level. The disparate findings may be explained by the fact that we used extremely high-intensity exercise in the initial 30-s that would have elicited a rapid increase in oxygen consumption in the active skeletal muscles; and this response may have resulted in a mismatch between oxygen demand and supply, thereby decreasing SpO₂. The mismatch, if any, could be exacerbated by minimal hyperventilation during the simulated 800-m run. Exercising at severe intensity causes hyperventilation such that $\dot{V}_E/\dot{V}CO_2$ increased relative to the lower-intensity domain. This hyperventilation could partly counteract EIAH as reported in previous studies (42). By contrast, $\dot{V}_E/\dot{V}CO_2$ during the simulated 800-m run remained low and was comparable to that observed during submaximal warm-up running, possibly limiting gas exchange, thereby widening the gap between oxygen demand and supply, reducing SpO₂.

There are alternative factors that could potentially contribute to the observed EIAH. Wetter et al. (41) postulated that both hyperthermia and reductions in pH may play a role in the development of EIAH during exercise at ~90% $\dot{V}O_{2max}$, by causing rightward shift of the hemoglobin–dissociation curve. Given the higher intensity of exercise used in our study, blood temperature and H⁺ concentrations would have more rapidly increased, potentially leading to an early and more pronounced

rightward shift of the hemoglobin–dissociation curve, thereby reducing SpO₂. Along this line, peak blood lactate concentration reached 17–19 mmol·L⁻¹ in the present study. Another possible factor that causes EIAH in the present study was expiratory flow limitation, indirectly evidenced by the plateau of $\dot{V}_E$ and V_T during the latter part of the simulated 800-m run. Furthermore, the occurrence of intrapulmonary arteriovenous shunting during exercise (44) may have also contributed to the development of EIAH in the present study. In this respect, Stickland et al. (44) reported that during high-intensity exercise, intrapulmonary shunt was associated with greater alveolar–arterial pressure difference for O₂, which is thought to be a major factor causing EIAH (41). However, it remains to be established whether very intense exercise, as used in the present study, exacerbates the influence of hyperthermia, acidosis, expiratory flow limitation, and/or intrapulmonary shunt on EIAH.

SpO₂ was not different between the caffeine and placebo trials, indicating that caffeine ingestion did not ameliorate severe EIAH. In line with this, although caffeine ingestion increased $\dot{V}_E/\dot{V}CO_2$ at the initial 10 s into the simulated 800-m run, this response was diminished subsequently. Moreover, $\dot{V}_E$ and $\dot{V}_E/\dot{V}O_2$ were comparable between the trials throughout the simulated 800-m run. Our results are inconsistent with previous studies reporting that caffeine ingestion increased $\dot{V}_E$ during running at 75%–100% $\dot{V}O_{2max}$ intensities in endurance-trained athletes (22,23). Although the precise reasons underlying the disparate findings are unknown, because uphill running protocol was used in the previous studies (fixed 18% or 2% increment every 2 min) (22,23) in contrast to only 1% inclination used in the present study, the effect of caffeine on exercise ventilation might not be clearly detected during level or near-level run by which most of middle- and long-distance running races are completed. Consistent with this, we observed no clear effect of caffeine consumption on $\dot{V}_E$ during the submaximal-intensity warm-up phase with a 1% incline, suggesting that the influence of caffeine on $\dot{V}_E$ remains unaltered across

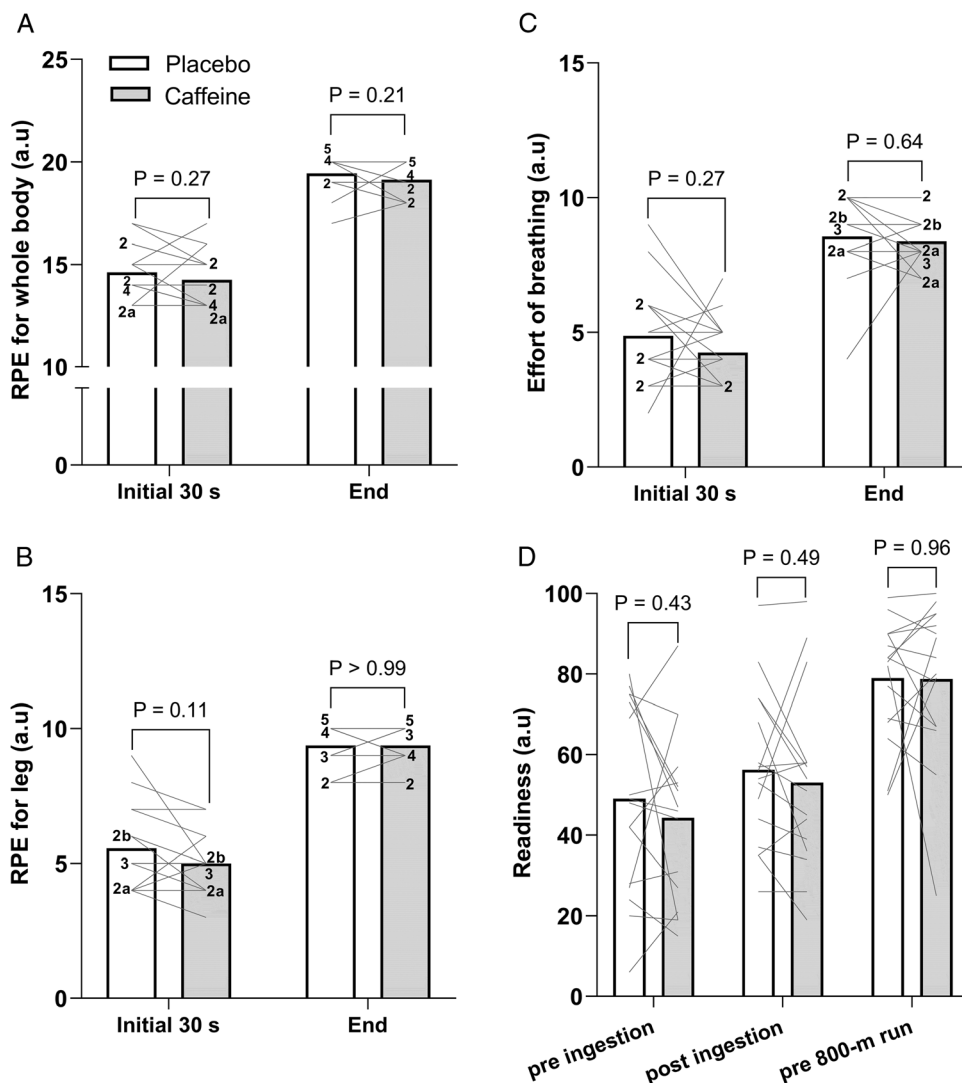


FIGURE 5—RPE for whole body (A) or leg (B), effort of breathing (C), and readiness for a simulated 800-m run (D) in both the placebo and caffeine trials. Both individual data and means are presented ($n = 16$). Data were analyzed using a paired t -test. Small numbers with or without alphabet placed at both placebo and caffeine line indicate the number of participants overlapped.

different exercise intensities. This is in accordance with a recent meta-analysis conducted by Glaister and Gissane (21), based on which a consistent increase in ventilation due to caffeine seems to be more pronounced during submaximal cycling but not running. Thus, it seems that the modality of exercise can modulate the impact of caffeine on exercise ventilation.

Without changing SpO_2 , $\dot{\text{V}}\text{O}_2$, and perceptions, caffeine ingestion improved the simulated 800-m run performance. The precise mechanisms responsible for the performance improvement resulting from caffeine ingestion remain undisclosed. Nevertheless, considering that caffeine ingestion increased exercise HR, it is conceivable that cardiac output may also have been increased. If this is the case, cerebral blood flow might have been increased, facilitating the delivery of oxygen to cerebral tissues without impacting SpO_2 , potentially enhancing performance in the simulated 800-m run. In this context, cerebral tissue deoxygenation seems to be a pivotal limiting factor in exercise performance (26).

The observed improvement in the 800-m run performance through caffeine ingestion could potentially be attributed to skeletal muscle mechanisms. Prior investigations have documented that caffeine intake (5–6 mg of caffeine per kg of body weight) led to an elevation in calcium release from the sarcoplasmic reticulum (19), motor-unit recruitment (16), and muscle fiber conduction velocity (15), all of which can increase muscle contraction. Given that caffeine ingestion may increase the force of contraction in fatigued, but not in fresh, muscles as proposed in a recent review article (13), it is plausible that caffeine ingestion might have amplified contractions of fast-twitch muscle fibers during the latter phase of the exercise, thereby enabling participants to sustain the run for an extended period. Alternatively, caffeine may have exerted a mitigating effect on the afferent signaling from the working muscles to the central nervous system, thereby reducing the inhibitory influence of locomotor muscle fatigue on the central motor drive, as previously hypothesized (13). Whatever the

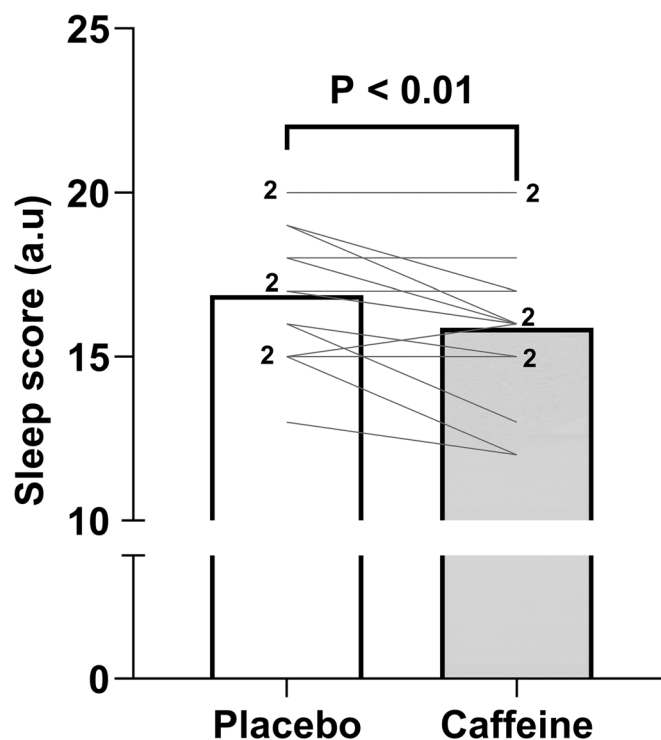


FIGURE 6—Sleep quality score evaluated next morning in both the placebo and caffeine trials ($n = 16$). Both individual data and means are presented ($n = 16$). Data were analyzed using a paired t -test. Small numbers placed at both placebo and caffeine line indicate the number of participants overlapped.

underlying mechanism(s), the prolonged exercise duration could have led to an enhancement in anaerobic metabolism, consequently resulting in an increase in the peak blood lactate concentration in the caffeine trial.

The results from the sleep questionnaire indicated that caffeine ingestion had a negative effect on subjective sleep quality, which is consistent with previous work that found caffeine ingestion at night (7:00 PM) can impair sleep quality (9). Although most of participants consumed caffeine in the morning (~9:00 AM) or early afternoon (~2:30 PM), the participants in the present study still experienced decreased sleep quality. Based on previous work (45), the large portion of elevated plasma concentration of caffeine seemed to be subsided before bedtime in the present study. Our findings indicate that, even if a significant amount of time has passed between caffeine ingestion and bedtime, caffeine can still affect sleep quality. Adequate sleep is crucial for the regulation of hormone secretion, restoration of metabolic process, and tissue function in athletes (46). Although research has shown that disrupted sleep before competition has not affected athletic performance, it can impair subjective feeling of fatigue, tension, and vigor (47). Given these findings, athletes may want to consider the potential impact of caffeine on sleep quality when competition in consecutive days.

Considerations. There are several considerations to note. First, although we simulated 800-m run in the present study, the performance was measured by time to exhaustion. Therefore, our results may not be simply applicable to actual 800-m run event. Second, we did not measure arterial blood gases, pH, and core temperature, all of which can influence EIAH. Future

studies are warranted to assess whether and how these factors modulate EIAH. Third, although the female participant's pattern of responses in the caffeine trial was similar to those of males, we cannot draw a firm conclusion regarding sex-related differences. Nevertheless, our main results were unchanged after excluding female participants' data. Thus, the impact of including female participant in our data analysis may be negligible. Fourth, we instructed the participants to memorize RPE and effort of breathing during the simulated 800-m run and subsequently report them after the exercise. However, we acknowledge that we cannot ascertain with certainty whether the reported RPE and effort of breathing accurately represent their subjective experience during the simulated 800-m run. Fifth, although we utilized a low dead space mask with a low-resistance breathing valve, we do not know whether wearing a mask contributes to EIAH occurring in the present study. Last, we evaluated the effect of caffeine on sleep solely through the utilization of questionnaires. Additional investigations using objective measures, such as sleep latency, sleep duration, and autonomic nervous system assessments, are warranted to delve more accurately into the influence of caffeine on sleep.

Practical applications. Our results suggest that caffeine may be an effective approach for enhancing performance in 800-m runs. However, it is important to keep in mind that caffeine may negatively impact subjective sleep quality, which could pose a concern for athletes participating in consecutive days of competition as discussed previously. As a result, it is essential for athletes to consider the impact that caffeine may have on their sleep quality when consuming it during such events.

CONCLUSIONS

We showed that caffeine intake (6 mg of caffeine per kg of body weight) improved exercise performance during simulated 800-m run exercise without altering severe EIAH, aerobic metabolism, and perceptions. Precise underlying mechanisms need to be assessed in the future.

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interest. The results of this study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of this study do not constitute endorsement by the American College of Sports Medicine.

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